香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

INTRODUCTION TO COMPUTER ENGINEERING: PROGRAMMING AND APPLICATIONS

TUTORIAL 1
INTRODUCTION

Ben Chen, SSE
2025

Outline

- Teaching fellows
- Schedule for tutorials
- Install Python
- About assignments and exams
- Some tips
- An example

Teaching fellow

- Instructor:
  - Prof ZHAO Junhua

Teaching fellows

➢Office Hour of TAs:

	Name	Email	Office Hour	Office Place	In charge of
白焰	BAI Yan	222015504@link.cuhk.edu.cn	Mon 11:00 - 12:00	Cheng Dao 204	Tutorial
朱一凡	ZHU Yifan	119010494@link.cuhk.edu.cn	Mon 15:00 - 16:00	Cheng Dao 320B	Tutorial
余天泽	Yu Tianze	223010076@link.cuhk.edu.cn	Mon 10:00-11:00	Cheng Dao 204	Tutorial
孟祥瑞	MENG Xiangrui	222010046@link.cuhk.edu.cn	Wed 11:00 - 12:00	Cheng Dao 318b	Tutorial, Assignment1
廖焕新	LIAO Huanxin	221019077@link.cuhk.edu.cn	Tues 16:00 - 17:00	Cheng Dao 320A	Tutorial, Assignment2
张一丹	Zhang Yidan	220019008@link.cuhk.edu.cn	Tues 15:00 - 16:00	Teaching C 412b	Tutorial, Assignment 3
郝一鸣	HAO Yiming	225010038@link.cuhk.edu.cn	Fri 14:00 - 15:00	Teaching C 412b	
孙正文泰	SUN Zhengwentai	224010153@link.cuhk.edu.cn	Wed 15:00 - 16:00	Teaching C 412b	
赵泓翔	ZHAO Hongxiang	224010208@link.cuhk.edu.cn	Tues 15:00 - 16:00	Teaching C 412b	
桂轩昂	GUI Xuanang	224010127@link.cuhk.edu.cn	Mon 10:00-11:00	Cheng Dao 204	Tutorial, Assignment 4

Timetable

- Arrangement of Tutorials:

Days	Tutorial Section	Time	Place	TA	Emial
Monday	T01	18:00 - 18:50	TC 109	余天泽，孟祥瑞，朱一凡 YU Tianze, MENG Xiangrui, ZHU Yifan	223010076@link.cuhk.edu.cn (YU)
Monday	T02	19:00 - 19:50	TC 109		222010046@link.cuhk.edu.cn (MENG)
Monday	T03	20:00 - 20:50	TC 109		119010494@link.cuhk.edu.cn (ZHU)
Tuesday	T04	18:00 - 18:50	TB 306	廖焕新，桂轩昂，白焰 LIAO Huanxin, GUI Xuanang, BAI Yan	221019077@link.cuhk.edu.cn (LIAO)
Tuesday	T05	19:00 - 19:50	TB 306		224010127@link.cuhk.edu.cn (GUI)
Tuesday	T07	20:00 - 20:50	TB 306		222015504@link.cuhk.edu.cn (BAI)
Wednesday	T06	18:00 - 18:50	TB 304	郝一鸣，张一丹，孙正文泰，赵泓翔 HAO Yiming, ZHANG Yidan, SUN Zhengwentai, ZHAO Hongxiang	225010038@link.cuhk.edu.cn (HAO)
Thursday	T09	18:00 - 18:50	TB 306		220019008@link.cuhk.edu.cn (ZHANG)
Thursday	T10	19:00 - 19:50	TB 306		224010153@link.cuhk.edu.cn (SUN)
Thursday	T08	20:00 - 20:50	TB 306		224010208@link.cuhk.edu.cn (ZHAO)

- The materials of tutorials can be downloaded from Blackboard.

Survey: How many of you have learned programming before?

What we do in tutorials?

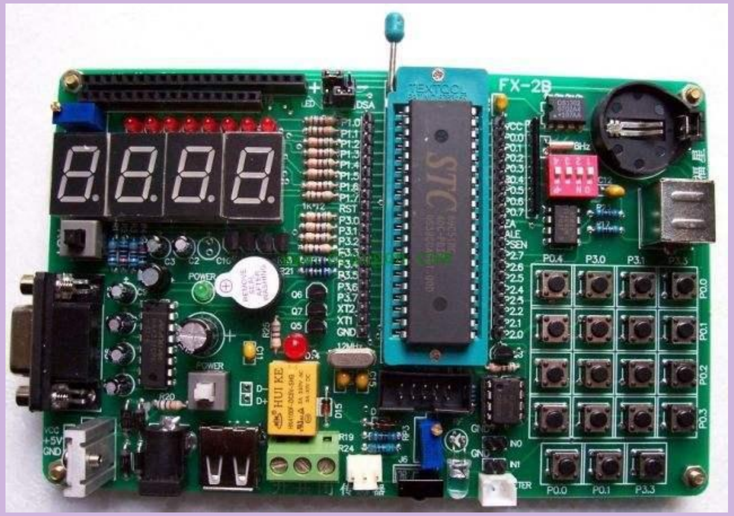
- Writing python codes to solve problems. Around 3~6 problems in one tutorial.
- TAs may explain the algorithms and python codes, or discuss and answer some questions from students about the lectures.

What is the computer programming?

- Computer programming is the process that professionals use to write code that instructs how a computer, application or software program performs.
- Code → Programs → Software / websites / apps ...
- What job can we take after learning computer programming?

Why we choose to learn Python?

- What is the computer programming language?
- Your code→ … → Machine instructions → Action.
- For example, suppose 00000000 represents "turn off all bulbs".
- So, why do we choose to learn Python?




Java
C
C++
PHP
JavaScript
Python
Kotlin
Rust
Visual Basic
Swift
Go
Dart
Forth
B
C#
D
Scala
TypeScript
Cobol
Objective-C
Groovy
Pascal
Logo
Ada
SQL
BASIC
Lisp
Tcl
ML
Tk
F#
Perl
Ruby
Prolog
Fortran
Julia
Lua
Eiffel
Scheme
Turbo Pascal
MATLAB
Erlang
Assembly Language

- Low Level Language and High Level Language.
- Assembly language is a low level language. It’s very difficult to use.
- C, C++, Python, Java, C#… are high level language.
- Executing efficiency VS coding efficiency.
- However, …
- Learning Python in this term helps you open a gate to the programming world. So grasp this opportunity!

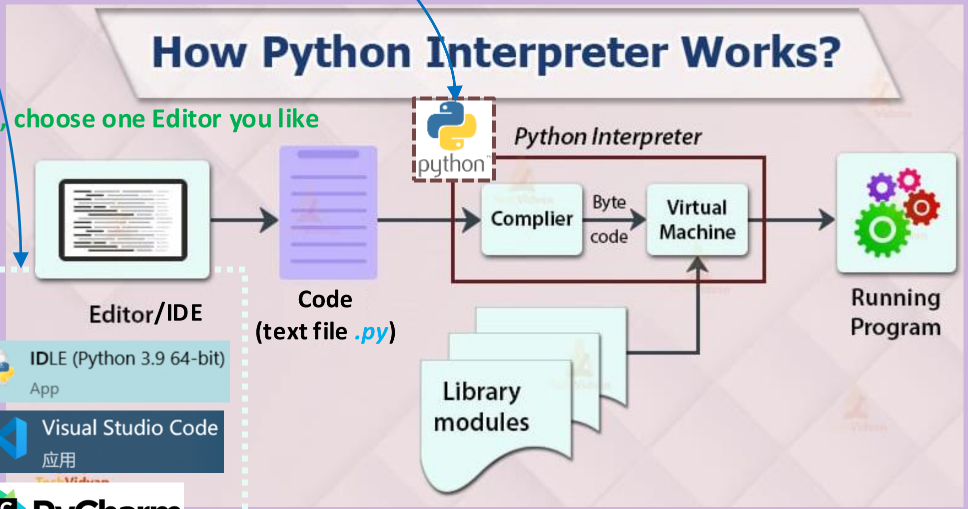
Python: Setup & First Run

Install Python, choose an IDE, and run your first program

- Python (**interpreter**) — runs your *.py* programs; CPython compiles to bytecode then executes.
- Editor/IDE — where you **write, run, and debug** code; examples include **VS Code**, PyCharm, and **IDLE** (bundled with Python and used for in-class demos).
- **Recommended start**: install **Python 3** and use **VS Code + Python extension** (you can still try small snippets in IDLE).

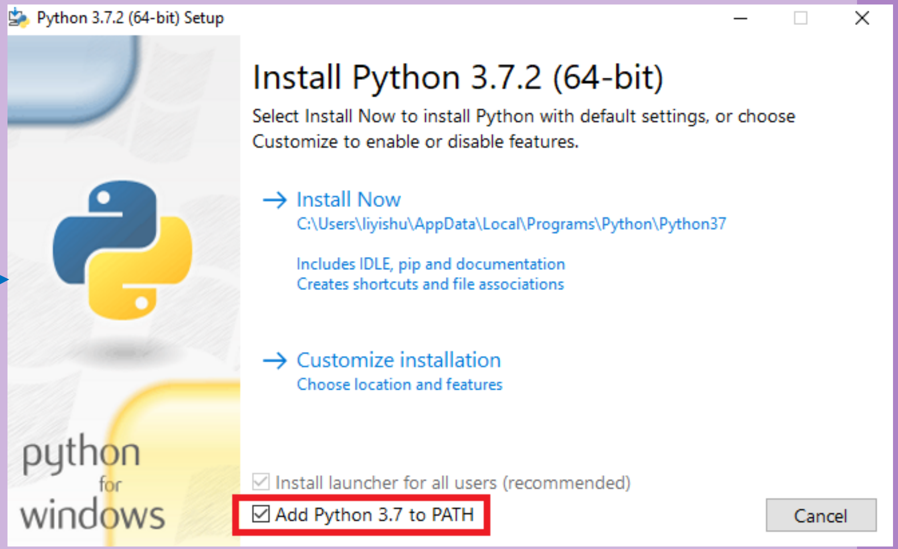
First, we need to download the interpreter!

Then, choose one Editor you like



Python Installation

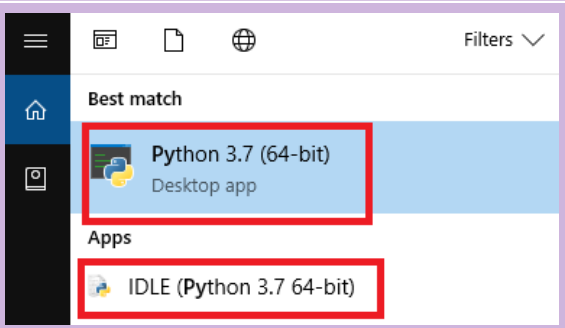
- Download the latest version (now 3.12) from https://www.python.org/
- Do remember to download the correct version for your own platform (Windows/Linux/MacOS)



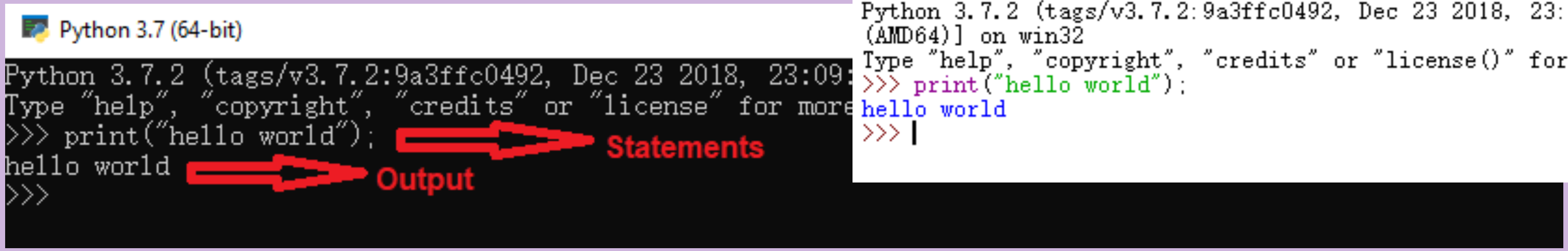
- Run installer → Install step by step:
  - CHECK "Add python.exe to PATH".
    (Make sure you've added Python path to environment variables)
  - Keep 'pip' and 'py launcher' selected (for Windows).

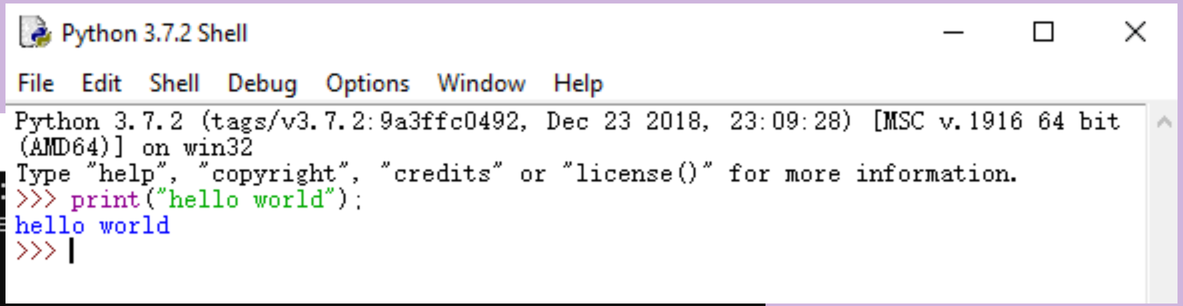
- Verify in Terminal (CMD/PowerShell):
  - py -V (version of launcher)
  - python --version (should show Python 3.x)
  - pip --version

How to write and run the program



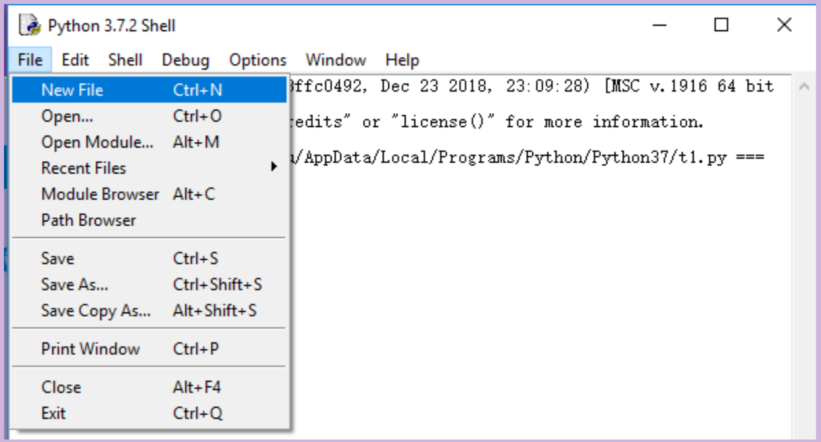
- Search "python" in start menu (Windows)
- Open Python desktop app or IDLE.
- You can write your code statement under shell (e.g. type "print("hello world");"
- Press enter and then the code will be executed. ("hello world" will be printed)

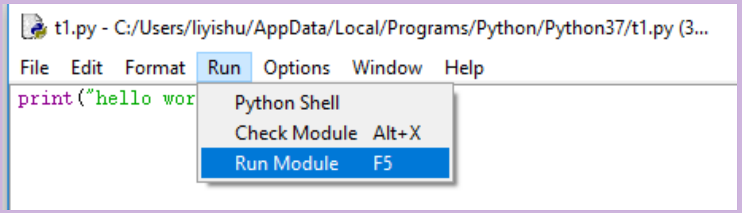


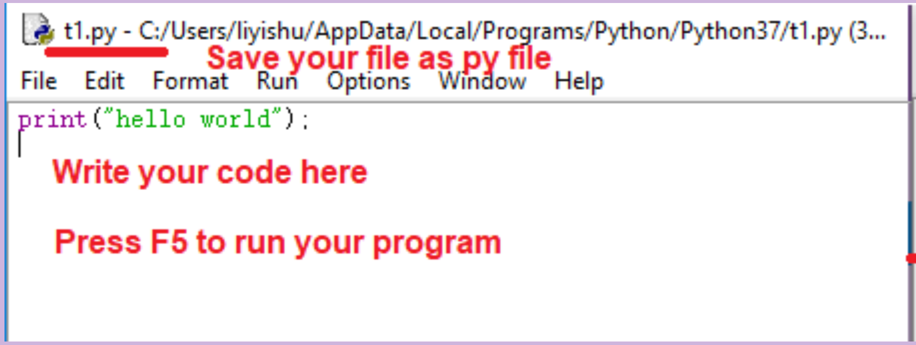


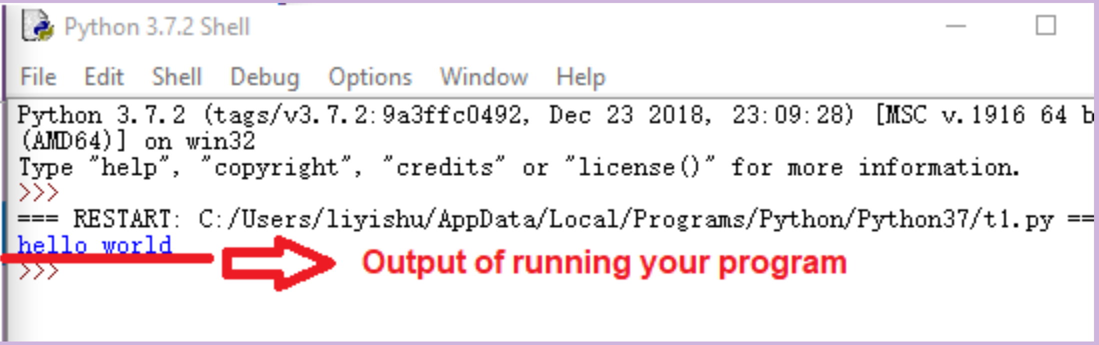
How to write and run the program

- In IDLE, you can also open or new a py file. Write your code in this file, and then press “F5” (or “Run” -> “Run Module”) to run your program.



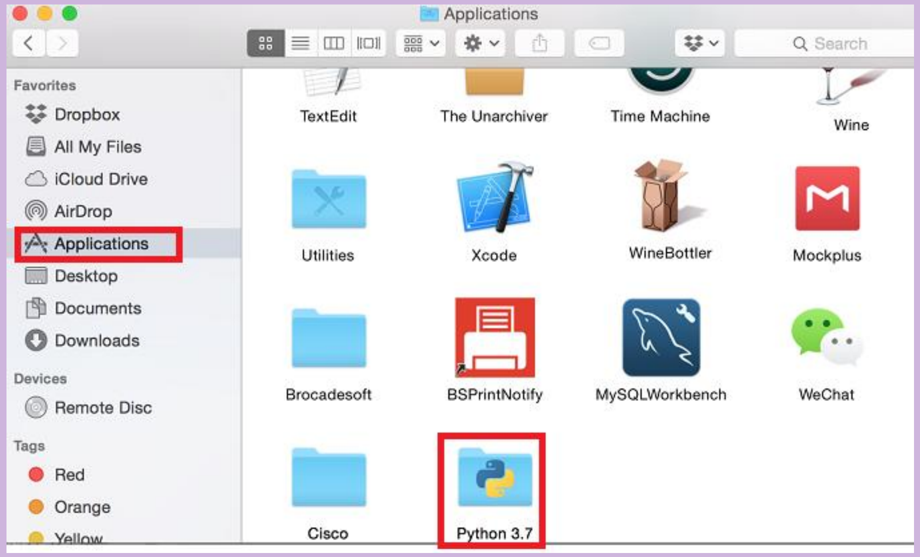


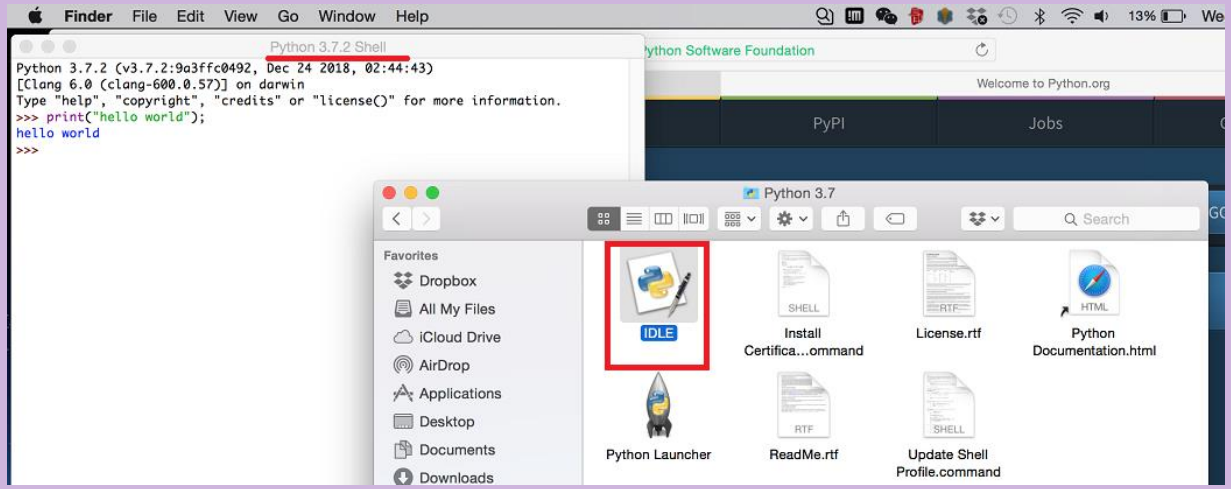




How to write and run the program

- For MacOS, you will find a folder named “Python 3.7” being displayed under Application.
- Double click “IDLE” and you will see the Shell and can write your py file or write your code in shell directly.





A better software…

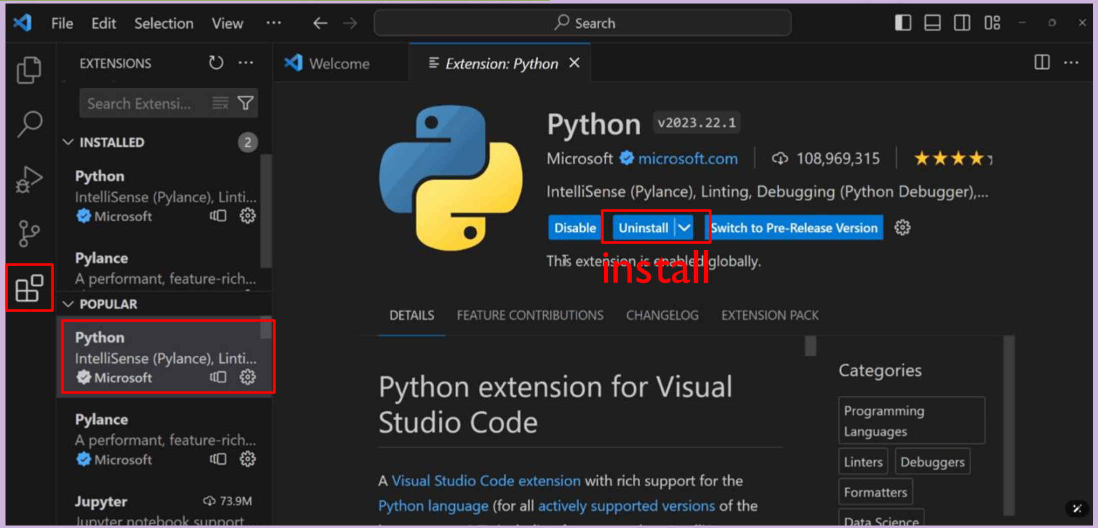
- Rather than utilize idle, I recommend you to download better IDEs.

(Idle is simple and great for first steps, but limited for bigger projects and add-ons.)

- IDEs help you manage your program projects.
- Some famous IDEs: VS Code, Spyder, Pycharm, …
- Show time…(Use VS Code as an example)

How to install Visual Studio Code

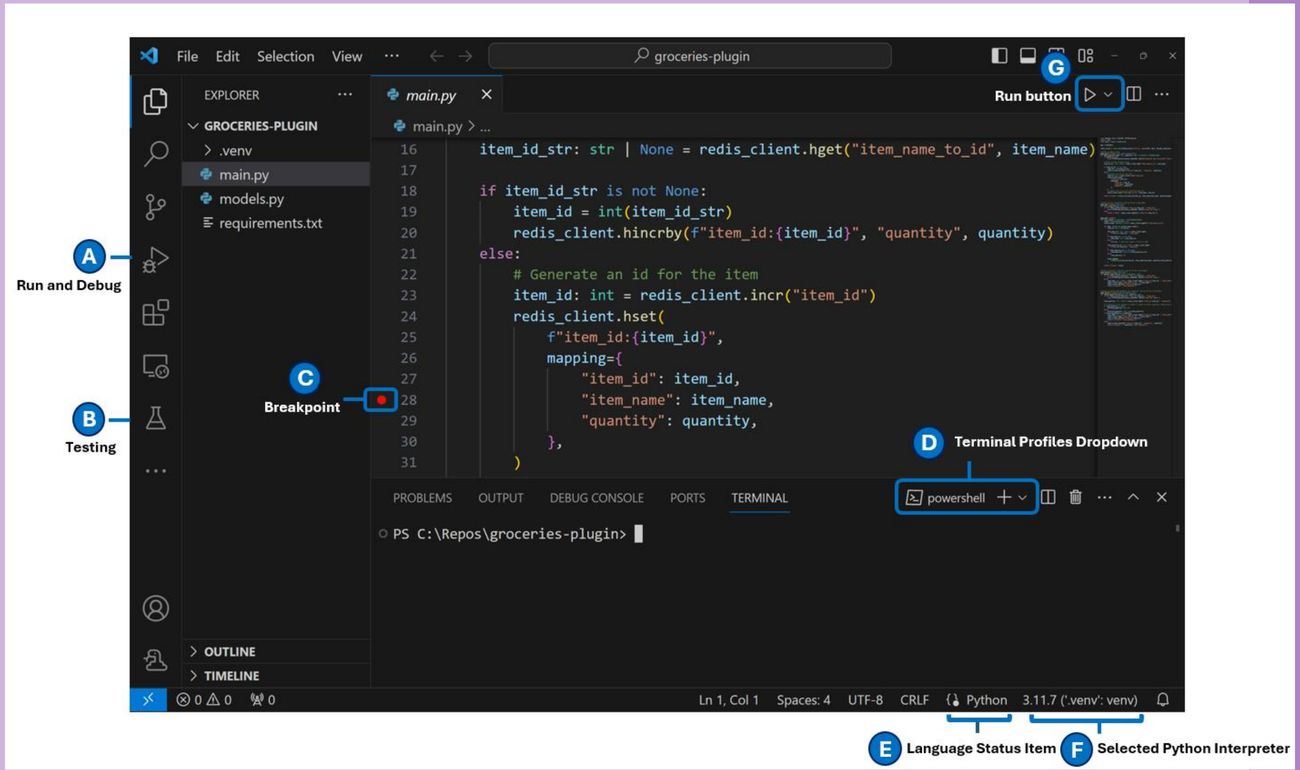
- Download and install VS Code for your OS via https://code.visualstudio.com/download
- Launch VS Code.



- Extensions (left sidebar) → search "Python" → install the one by Microsoft.
- Done. (Attention: Interpreter ≠"Extension" there, Python 3 was already installed last step.

Quick start — Open Python in VS Code and run

- Open folder: File → Open Folder… (Create one). Check Language Status (E) shows Python.
- Select interpreter: click the status-bar interpreter (F) and choose your Python or .venv.
- Create xxx.py with: print("Hello, VS Code")
- Click ▶ "Run Python File" to see output in the terminal panel.
- Switch shells via Terminal Profiles (D) if needed.
- Optional debug: set a Breakpoint (C) by clicking the gutter. Open Run and Debug (A) → Python File to step through.



Reference links: https://code.visualstudio.com/docs/python/python-quick-start?originUrl=%2Fdocs

knowledge points of Python

- identifier
- basic grammars
- variables’ type
- operator
- conditional statement, loop statement
- string
- list, tuple, dictionary
- function

knowledge points of Python

- module
- file, I/O
- OOP: Object-oriented Programming

Key point

- In order to improve your programming skill, you must try your best to write more code.
- Write comments when coding. The volume of comments need to cover over 1/3 of your codes.

About assignments and exams

- Assignments are about writing codes to solve problems, codes (.py) and explanations(.pdf) about your programs(zipped into one file) are supposed to be uploaded to the blackboard system before the deadline.
- Passing deadline will deduct some points, three days after deadline won't be accepted.
- Exams are paper forms, about understanding the lecture notes and writing codes on exam papers.

Some tips

- Learn based on lecture notes. Practice, practice and practice.
- Programs are to help you solve problems, but it is still you to solve the problems, computer is just to calculate faster. So you should solve the problems by yourself first for the simpler cases, then design algorithms to let computer do it.
- Before midterm: basic tools(data types, if else, for, while) to solve problems, similar to a designed calculator.
- After midterm: abstract concepts(functions, class, algorithm, list, tree etc.), to combine tools into a "machine" with functionalities.

An example: get email addresses

- Problem:

  Once I need to send an email to a large amount of students, but I only get a file with their student ids. How can I get a file with their email addresses.

- Solution:

  To solve it, just import the id file and add “@link.cuhk.edu.cn” after that, and write into a new file.

An example: get email addresses

- Tips for writing codes:
  1) Use words to represent variables in order to remember their meanings easily;
  2) Write comments using "#" to explain important parts of your algorithm.

Announcement

Remember to bring your laptop next time!

Any questions or suggestions?